Final Report for Mid Term Project

RSH E Shopping Junction

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Learning Outcomes:

- In this project we learnt scrapping data from websites, using requests.
- We learnt to store data in csv files using pandas.
- We learnt how to design UI (User Interface) and connect it with the backend to work properly.
- We learnt implementing multiple functions on data like, sorting, Multiple level Sorting, searching.

Executive Summary:

The software includes data that can be organized through various sorting algorithms and searched using different search algorithms. This enables users to view the entire product catalog and arrange the products in accordance with their specific requirements. There are a total of eight sorting algorithms provided, allowing users to choose any of them for sorting purposes.

Within the software, there are nine product attributes that users can choose for sorting. Users are required to select both the sorting algorithm and the specific attribute they wish to use for sorting. Additionally, the software offers a search function that allows users to find products by simply entering relevant keywords or terms.

Scrapping:

- **Scrapping Link:**

"https://www.ebay.com/sch/i.html?_from=R40&_trksid=p2334524.m570.l1313&_nkw=printer&_sacat=0&LH_TitleDesc=0&_fsrp=1&Brand=Zebra%7CLexmark&rt=nc&_odkw=printer&_osacat=0&_dcat=1245&_oaa=1"

1. **Problem Statement:**
   In contemporary times, individuals embarking on a new business venture must possess a comprehensive understanding of the products they intend to sell. They require access to a platform that can provide them with all the necessary information about these products.
2. **Project Overview:**
   This software has been designed to address the issue mentioned in point 1. It assists users in acquiring product information, including user ratings, enabling them to identify highly-rated products to stock. This, in turn, can enhance their earnings. Furthermore, the software allows users to assess the prices of scraped products, helping them select items within their budget for their shop.

   The website continuously updates with newly launched products in the market, offering users the opportunity to be among the first to stock and sell these items, potentially leading to substantial profits. Additionally, the software provides users with the ability to check

shipping rates for products they wish to source from the website, allowing them to choose the most suitable seller..

Attributes:

- Type Of Product
- Name Of Product
- Brand Name
- Out Of Stock
- Price Of Products
- Discount On Product
- Delivery Charges
- Sold Products
- Ratings

Algorithms:

1. **For Sorting:**
   - Merge Sort
   - Insertion Sort
   - Selection Sort
   - Bubble Sort
   - Quick Sort
   - Counting Sort
   - Radix Sort
   - Bucket Sort
2. **For Searching:**
   - Linear Search

3. Obstacles:

Certain websites have restrictions that prevent data scraping. The website we're currently utilizing permits data scraping for up to five pages. However, when we accessed a link where the page structure was altered, and new products were incorporated into existing pages, the software encountered issues. Specifically, when dealing with substantial data, such as when working with thousands of products, the software experiences slowdowns, and the sorting process becomes time-consuming.

4. Resources for development:

- Spyder as IDE
- Python as Programming Language
- PYQT5 for the Designing of UI

Description for Algorithms:

Merge Sort:

Merge sort is a comparison-based algorithm. It works on the divide and conquer principle. It divides the array into small parts until there are only two elements left. Then it swaps the elements and sorts them. When all the elements are sorted it again merges them to an array.

Time Complexity:

Worst Complexity: O(n*log(n))

Average Complexity: O(n*log(n))

Best Complexity: O(n*log(n))

Strength:

It is faster while working with the large amount of data.

Weakness:

It is slower while working for the small amount of data.

Insertion Sort:

Insertion sort is a comparison-based algorithm in this algorithm array is divided into two parts sorted elements are put on one side and the unsorted on the other side. The algorithm keeps finding the elements from the unsort part and put it in the sorted part of the array.

Time Complexity:

Worst Complexity: $O(n^2)$

Average Complexity: $O(n^2)$

Best Complexity: O(n)

Strength:

It works faster for the small amount of data.

Weakness:

It works slower if the amount of data is large.

Selection Sort:

Selection sort is a comparison-based algorithm it finds the smallest number and then takes it to the first index of the array then finds the second smallest number and put it to the second index and similarly for others. In the last array is sorted.

Time Complexity:

Worst Complexity: $O(n^2)$

Average Complexity: $O(n^2)$

Best Complexity: $O(n^2)$

Strength:

No additional temporary memory storage is need for the selection sort. It sorts the elements in the same array.

Weakness:

With the increase in the amount of the data it becomes slower.

Bubble Sort:

Bubble sort is a comparison-based algorithm It compares two adjacent elements and swap them then same for the next two elements. It does this process until all the elements are sorted in the array.

Time Complexity:

Worst Complexity: $O(n^2)$

Average Complexity: $O(n^2)$

Best Complexity: $O(n)$

Strength:

Its code is very small and easy to understand. It is faster for the small amount of data and for the data that is mostly in order.

Weakness:

It takes a lot of time for sorting large amount of data. It can take $O(n^2)$ time.

Quick Sort:

It is a comparison-based sorting algorithm. It has a pivot point. In this algorithm the array is broken into small parts then it makes comparison of the elements around the pivot and arranges the elements. In the last the array is again combined. The resulted array is sorted now.

Time Complexity:

Worst Complexity: $O(n^2)$

Average Complexity: O(n*log(n))

Best Complexity: O(n*log(n))

Strength:

Quick Sort is very fast in sorting the elements. As it has to decide from the pivot point to sort the elements.

Weakness:

Quick sort is recursive there for it can take $O(n^2)$ time to sort the elements in the worst case.

Bucket Sort:

It divides items into multiple groups called buckets. Bucket sort items are first divided evenly into groups called buckets and then sorted using other sorting algorithms. Then sort and collect the elements.

Strength:

It works faster on float value.

Weakness:

Can not work faster on any other data type except float.

Time Complexity:

Best Case: $\Omega(n +k)$ Average: $\theta(n +k)$ Worst case: $O(n^2)$

Radix Sort:

It is a linear sorting algorithm in which we sort elements of arrays by comparing their digits. First, very right digit of every elements of array is compared with each other and sort elements with respect to very right digit. This process repeats for further digits and hence array becomes sorted.

Time Complexity:

O(d*(n+b))

Strength:

Radix Sort has linear time complexity when the number of digits in the input numbers is relatively small compared to the number of elements.

Weakness:

Radix Sort is not a comparison-based sorting algorithm, which means it is not suitable for sorting elements with arbitrary order relationships.

Counting Sort:

Counting Sort is a linear algorithm that sorts the elements of and array by counting the number of occurrences of each unique element in the array. If searches for the occurrences of a number and then put it in the array at that index of number. For example, the number searched is 5 then it will put all the occurrence of the number at index five of the array.

Time Complexity:

Worst Complexity: O(n)

Average Complexity: O(n+k)

Best Complexity: O(n+k)

Strength:

Counting sort is the faster then all the comparison bases sorting algorithms.

Weakness:

Counting sort is only works for the integers. It sorts the string because it cannot make buckets of characters to strings there. It need integers to keep the count of the elements.

Searching Algorithms:

Linear Search:

In this searching algorithm. The element is searching is started from the first index of the array. If the element is found before the last elements it returns the elements. If not, it will run until the last index of the array.

Time Complexity:

Worst Complexity: O(n)

Average Complexity: O(n)

Best Complexity: O(1)

Strength:

The biggest advantage of the linear sort is that it can find the element even from the unsorted array. It works very good for small amount of data. It can find strings from the data also.

Weakness:

For large amount of data linear sort may take a lot of time to search an element.

GUI of the Software:

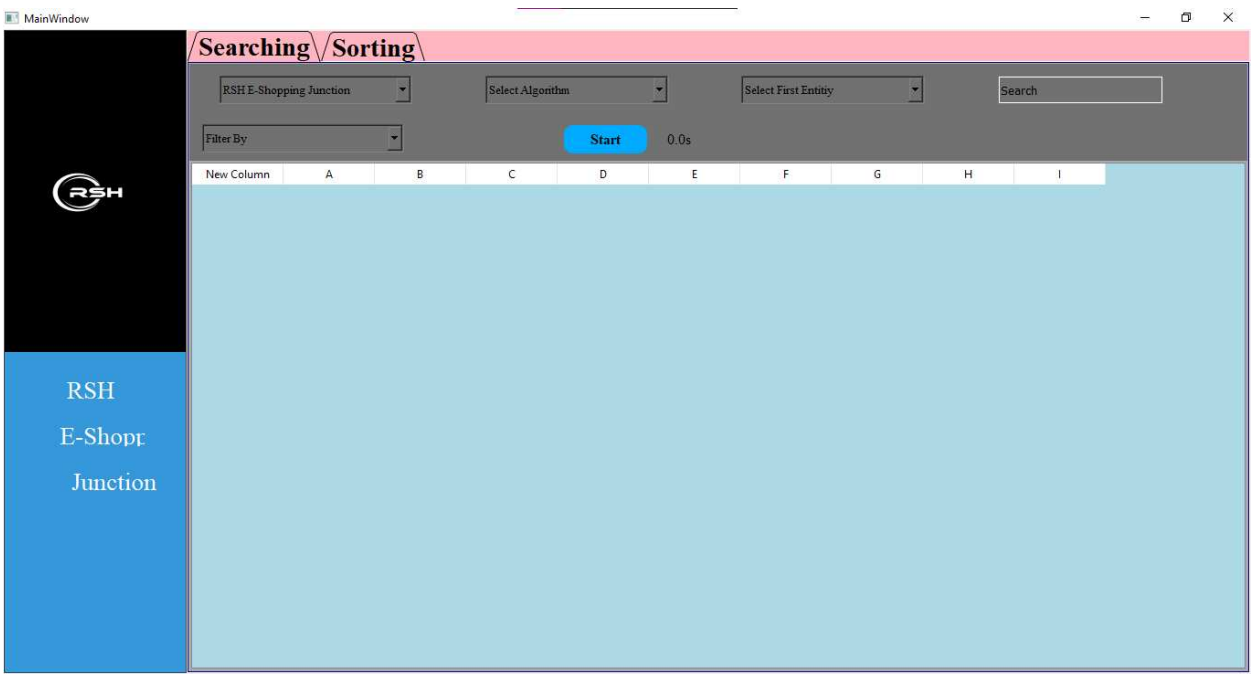

Fig 1: Basic Interface

	Type Of Product	Name Of Product	Brand Name	Out Of Stock	Price Of Products	scount On Produ	Delivery Charges	Sold Products	Ratings
1	Air pods	Shop on eBay	on	Available	20.0	0	0.0	0	0.0
2	Air pods	Apple AirPods ...	AirPods	Available	199.0	0	22.51	253	4.5
3	Air pods	Apple AirPods ...	AirPods	Available	100.0	0	35.88	0	4.5
4	Air pods	Apple AirPods ...	AirPods	Available	154.0	0	26.8	0	4.5
5	Air pods	Apple AirPods ...	AirPods	Available	45.0	0	27.88	1004	4.5
6	Air pods	EMPTY BOX ...	BOX	Available	20.0	0	31.57	0	0.0
7	Air pods	AirPods Pro 2n...	Pro	Available	149.0	0	16.8	0	0.0
8	Air pods	Airpods	Airpods	Available	18.0	0	31.51	0	0.0
9	Air pods	New ListingAp...	ListingApple	Available	13.0	0	31.41	0	5.0
10	Air pods	Apple AirPods ...	AirPods	Available	9.9	0	20.12	44	0.0
11	Air pods	Apple Air Pods ...	Air	Available	49.0	0	26.66	0	4.5
12	Air pods	AUTHENTIAppl...	Apple	Available	48.0	0	28.98	0	4.5
13	Air pods	Air Pods Pro 2n...	Pods	Available	46.0	0	26.26	0	0.0
14	Air pods	Apple AirPods ...	AirPods	Available	85.0	0	27.29	0	4.5
15	Air pods	Apple Airpods ...	Airpods	Available	79.0	0	10.12	0	4.5
16	Air pods	Apple AirPods ...	AirPods	Available	119.0	0	28.4	0	4.5
17	Air pods	airpods pro	pro	Available	57.0	0	5.25	0	0.0
18	Air pods	Apple AirPods ...	AirPods	Available	75.0	0	80.0	232	4.5
19	Air pods	*FAULTY ...	UNBOXED*	Available	30.0	0	39.33	0	4.5

Fig 2: Sorting Interface

	Type Of Product	Name Of Product	Brand Name	Out Of Stock	Price Of Products	scount On Produ	Delivery Charges	Sold Products	Ratings
1	Air pods	Shop on eBay	on	Available	20.0	0	0.0	0	0.0
2	Air pods	Apple AirPods ...	AirPods	Available	199.0	0	22.51	253	4.5
3	Air pods	Apple AirPods ...	AirPods	Available	100.0	0	35.88	0	4.5
4	Air pods	Apple AirPods ...	AirPods	Available	154.0	0	26.8	0	4.5
5	Air pods	Apple AirPods ...	AirPods	Available	45.0	0	27.88	1004	4.5
6	Air pods	EMPTY BOX ...	BOX	Available	20.0	0	31.57	0	0.0
7	Air pods	AirPods Pro 2n...	Pro	Available	149.0	0	16.8	0	0.0
8	Air pods	Airpods	Airpods	Available	18.0	0	31.51	0	0.0
9	Air pods	New ListingAp...	ListingApple	Available	13.0	0	31.41	0	5.0
10	Air pods	Apple AirPods ...	AirPods	Available	9.9	0	20.12	44	0.0
11	Air pods	Apple Air Pods ...	Air	Available	49.0	0	26.66	0	4.5
12	Air pods	AUTHENTIAppl...	Apple	Available	48.0	0	28.98	0	4.5
13	Air pods	Air Pods Pro 2n...	Pods	Available	46.0	0	26.26	0	0.0
14	Air pods	Apple AirPods ...	AirPods	Available	85.0	0	27.29	0	4.5
15	Air pods	Apple Airpods ...	Airpods	Available	79.0	0	10.12	0	4.5
16	Air pods	Apple AirPods ...	AirPods	Available	119.0	0	28.4	0	4.5
17	Air pods	airpods pro	pro	Available	57.0	0	5.25	0	0.0
18	Air pods	Apple AirPods ...	AirPods	Available	75.0	0	80.0	232	4.5
19	Air pods	*FAULTY ...	UNBOXED*	Available	30.0	0	39.33	0	4.5

Fig 3: Searching GUI